

Detailed Trading Requirements Spec

Part 1: Multi-Agent Architecture, Entry Mechanics & System A

1. Detailed Multi-Agent & CEO Decision Flow

The core system runs as a Multi-Agent Orchestrator. It delegates data gathering and localized calculations to distinct agents. This division prevents single API failures from freezing the system. The CEO Agent is the sole decision processor. It combines reports from System A (Technical) and System B (Institutional) to calculate consensus trades.

Entry Scenarios Handling

How entries are handled exactly:

1. Signal Calculation: At the exact close of each 5m candle, the Technical Agent computes a weighted score (0-100). Simultaneously, the Institutional Agent checks order flow and volume delta to assign a Neutral/Buy/Sell rating.
2. Trend Continuation Entry: Triggers if System A score is ≥ 80 AND System B is BUY. The CEO marks this as Trend Continuation. The stop-loss is set at $1.5 \times \text{ATR}$ below entry. Target 1 and Target 2 are both projected. This represents high institutional confirmation of chart momentum.
3. Momentum Scalp Entry: Triggers if System A score is 65-79 (regardless of System B) OR if System A is ≥ 80 while System B is Neutral. This represents a fast trade opportunity. The position size is halved, stop-loss is tight ($1.0 \times \text{ATR}$), and Target 2 is disabled (auto-exits fully at Target 1).
4. Entry Zone Execution: CEO outputs an exact price range: $[\text{Close Price}, \text{Close Price} + 0.25\% \text{ ATR}]$. This prevents chasing breakouts when the spread widens.

Conflict Management Logic

If System A indicates a strong buy (≥ 80) but System B (Institutional Flow) registers a Sell verdict (meaning cumulative delta or DOM ask depth indicates aggressive selling), the CEO Agent blocks the trade. Instead of silence, the system generates a 'Conflict Report' detailing the mismatch. This info-only alert is sent to Telegram and logged to `conflict_log.jsonl` to allow weekend validation.

2. System A Indicators - Formulas & Rationale

Here we detail the exact mathematical formulas for the 7 System A indicators and why institutions rely on them:

1. VWAP (Volume Weighted Average Price)

Formula: $\text{VWAP} = \text{Sum}(\text{Price} * \text{Volume}) / \text{Sum}(\text{Volume})$ recalculated from market open.

Why: Moving averages weight all periods equally, ignoring volume. VWAP reflects true institutional fair value. Algorithms execution desks benchmark their trade costs against VWAP. Entering at a VWAP bounce aligns with institutional accumulation.

2. EMA 9/21 Ribbon

Formula: $\text{EMA}_t = \text{Price}_t * \alpha + \text{EMA}_{\text{prev}} * (1 - \alpha)$, where $\alpha = 2 / (N + 1)$.

Why: EMAs weight recent prices more heavily than SMAs, responding faster to intraday momentum. The crossover of the 9 EMA above the 21 EMA signals short-term acceleration, while the expansion of the distance between them measures momentum velocity.

3. MACD Histogram Slope

Formula: $\text{MACD_Line} = \text{EMA}(12) - \text{EMA}(26)$; $\text{Signal_Line} = \text{EMA}(\text{MACD}, 9)$; $\text{Hist} = \text{MACD_Line} - \text{Signal_Line}$.

Why: Rather than just looking at cross-overs, we measure the slope of the Histogram over 3 candles. A rising histogram slope ($\text{Hist}[0] > \text{Hist}[-1] > \text{Hist}[-2]$) confirms that momentum is actively expanding before a cross happens.

4. RSI Hidden Bullish Divergence

Formula: $RSI = 100 - (100 / (1 + RS))$, where $RS = \text{Average Gain} / \text{Average Loss}$ over 14 bars.

Why: While retail traders search for simple oversold values ($RSI < 30$), quants search for Hidden Bullish Divergence. When price makes a higher low but RSI makes a lower low, it indicates structural strength and that selling pressure is exhausted.

5. Volume Delta & Confirmation

Formula: Current Volume $> 1.5x$ of the 20-period moving average volume.

Why: Price action without volume is algorithmic noise. High-volume confirmations ensure that institutional block orders are supporting the breakout or support bounce.

6. Bollinger Band Squeeze

Formula: Bollinger Bands (20, 2) compress inside Keltner Channels (20, 1.5).

Why: Markets cycle from low volatility to high volatility. The BB Squeeze identifies periods of volatility compression. When the bands expand outward, it indicates a high-probability breakout event.

7. Order Block Proximity

Formula: The price zone of the last down-close candle before the initial breakout rally.

Why: Institutions accumulate size in blocks. When price returns to these historical accumulation zones, limit orders are triggered, creating highly reliable support levels.

3. Data Feeds & Emergency Price Guardian

IBKR WebSocket Feeds

To handle 10 concurrent watchlist assets without bottlenecking the CPU or getting rate-limited:

- Bar Stream: Subscribes to 5m real-time bar updates via the TWS API.
- Tick Stream: Subscribes to raw tick-by-tick data to compute real-time volume calculations.
- Queue Staggering: Queries options chains and HTF bars via an internal queue, allowing only 6 API queries per 10 seconds.

Emergency Price Guardian (Real-time Crash Protection)

Because waiting 5 minutes for a candle to close during a crash is highly risky, a separate thread runs the Price Guardian:

- Runs a 5-second loop querying the latest tick price for watchlist tickers.
- Compares price directly against the locked stop-loss.
- If price falls below the stop-loss, it issues an immediate emergency exit message to Telegram and marks the position as closed.
- If price drops $> 1.5\%$ within 3 minutes, it triggers a flash crash freeze, halting all buy signals for that asset.